

Green Cycle - Final Project

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1. Introduction

This is the final document for the Green Cycle project, completed as part of the IT Management course during the academic year 2025/2026. It summarizes all work produced throughout the course and reflects on how each phase connected to the next.

Green Cycle is a web platform for sharing and discovering upcycling projects. The core problem it addresses is that upcycling information is scattered across Reddit, Instagram, Pinterest, and YouTube with no consistent structure or quality. Beginners in particular have no reliable starting point. The platform brings this content together with structured project submissions, material-based filtering, and basic social features like comments, private messaging, and achievement badges.

The target groups are eco-conscious individuals, teachers, and small eco-businesses, each with distinct needs that were explored through dedicated research before a single line of code was written.

2. Project Charter

The project charter was the first document and served as the formal contract for all work that followed. It locked in the scope, timeline, budget, and technology decisions so that later phases had a clear reference point.

The platform was scoped to cover user registration and profiles, project posting with structured materials and instructions, browsing and filtering, commenting and private messaging, content moderation, an admin panel, and an achievement system. The technology stack: React, Node.js, PostgreSQL was specified by the client, and all chosen tools are free and open-source.

The project runs for four months, divided into four phases with fixed milestones: backend architecture by end of March, primary features by end of April, secondary features by end of May, and final delivery by end of June. The budget is €3,000 for the development team, with hosting costs covered separately.

Three risks were identified upfront: scope creep (rated medium likelihood, high impact, mitigated by treating the feature list as contractually locked), performance issues (low likelihood, medium impact, addressed through pre-delivery load testing), and security vulnerabilities (low likelihood, high impact, addressed through strict code review). These were not written as formalities; the scope creep risk in particular was referenced later in the stakeholder analysis when defining how to handle client-side expectations.

3. Personas and Problem Scenarios

Before any user research or feature design, three personas were built to represent the platform's target groups. The goal was to ground every subsequent decision in a specific person's context rather than abstract assumptions.

Andrew is a 28-year-old graphic designer who wants to upcycle things at home but keeps getting lost in scattered online content. His key frustrations are that guides assume prior craft knowledge, that there is no way to search by the specific materials he already owns, and that posting his work anywhere general gets no meaningful response.

Tim is a 44-year-old high school teacher running an after-school Green Club. He spends more time searching for suitable project ideas than actually running them. He also has a folder of tested classroom activities with no good place to share them.

William is a 35-year-old entrepreneur who turns industrial fabric scraps into accessories. He wants to find eco-conscious customers without a marketing budget, and to source recyclable materials without relying entirely on personal contacts.

These three personas shaped every phase that came after: the hypotheses tested, the user stories written, and the stakeholders prioritized.

4. User Stories

With the personas defined, user stories translated each persona's needs into concrete, testable requirements. Each persona received three user stories, and each user story was broken down into test cases that serve as acceptance criteria for implementation.

Andrew's stories cover finding beginner-friendly projects by material and difficulty, following a structured step-by-step guide on a project page, and sharing completed projects to receive community feedback.

Tim's stories cover finding classroom-appropriate projects using age group and cost filters, contacting a project author via private message, and publishing his own tested classroom activities.

William's stories cover building a public profile with visible achievements and published projects, reaching an eco-conscious audience through detailed project posts, and finding material suppliers within the community.

5. Project Stakeholders

The stakeholder analysis was conducted after the user stories were defined, at a point when the team had a clear enough picture of the product to assess who has an interest in it and where tension could arise.

The most directly involved stakeholders are the client (the IT Management course), the development team itself, and the three user groups corresponding to the personas. Two secondary stakeholders are platform moderators and administrators, whose needs are less visible during development but critical for long-term operation.

6. Hypothesis Testing

Before committing to any development, the core value assumptions behind the platform were tested through direct interviews with potential users.

The findings confirmed two of the three hypotheses outright. The eco-conscious individual interview confirmed that material-based search is the single most wanted missing feature, the participant described having a bag of old materials and no way to search by what she already owned. The teacher interview confirmed the time problem directly. He described spending more time finding an activity than preparing the actual lesson, and noted having a folder of tested classroom activities with nowhere meaningful to share them.

The third hypothesis was: small eco-business owners cannot reach a relevant audience through general social media. The hypothesis was marked as only partially confirmed. The reach problem was clear, but the participant raised a concern about whether the material sourcing feature would actually be useful in practice.

Based on these findings, all required changes were made to the prototype, resulting in better fulfillment of user needs on the platform.

7. Usability Testing

After testing and updating the prototype, usability testing was conducted to evaluate whether the interface actually supports the user stories in practice.

Five tasks were designed, each mapped to a specific user story: finding a project by material and difficulty, reading a project page, finding a classroom-appropriate project, submitting a new project, and contacting a project author via private message.

The clearest usability issues appeared in Task 3 and Task 4. The classroom filter and estimated project cost were difficult to notice, confirming concerns raised during stakeholder analysis, while the project submission flow caused problems for all

participants, especially the easily missed “Published” checkbox and unclear materials search field. In contrast, Task 2, reading a project page, proved intuitive across all subjects, which was encouraging given that it represents one of the platform’s most frequently used flows.

8. Scrum Basics and Engineering Management

The research and planning phases defined what to build, while Scrum basics and engineering management defined how to build it during the four-month period. Boris acted as Product Owner and backend developer, Janko as Scrum Master and backend developer, and Nikola as frontend developer. The team used two-week sprints, Fibonacci story points, and GitHub Projects for lightweight Scrum management.

Sprint 1 focused on the backend foundation, including the database schema, authentication, role-based access control, and frontend login and registration pages. Work was assigned according to team roles and expertise rather than equal distribution. Sprints 2-4 were planned at a higher level, covering core features, secondary features, and finally testing, deployment, and documentation.

From an engineering management perspective, the team avoided micromanagement, resolved blockers proactively through the Scrum Master role, and assigned tasks based on actual skills instead of artificial equality.

9. Final Application Showcase

Register form

Tom

tom@gmail.com

Choose file

No file chosen

Register

GreenCycle

Search...

Filters

Apply

new

 Logout

Add New Project

Eco-Friendly Gift Wrapping Kit

Create a reusable, eco-conscious gift wrapping kit using old newspapers, string, and other recycled materials. Ideal for birthdays, holidays, or any occasion – stylish and sustainable! 🌱

Art

Medium

60-90

Optional: Add a couple of plastic bottle caps as decorative elements or stamps.

Store everything in a small box or envelope made from reused cardboard.

☐ Published

Thumbnail:

Choose file

No file chosen

Images:

Choose files

No file chosen

Search Materials

GreenCycle

Search...

Filters

Apply

new

Logout

Edit Project

Eco-Friendly Gift Wrapping Kit

Create a reusable, eco-conscious gift wrapping kit using old newspapers, string, and other recycled materials. Ideal for birthdays, holidays, or any occasion – stylish and sustainable!

Art

Medium

60-90

Select clean, visually interesting pages from old newspapers to use as wrapping sheets.

☒ Published

Thumbnail:

Choose file No file chosen

Images:

Choose files No file chosen

Search Materials

GreenCycle

Search...

Filters

Apply

new

Logout

Bottle Cap Mosaic Art



Category: art
Difficulty: easy
Time: 120 - 180 min

T-Shirt Tote Bag



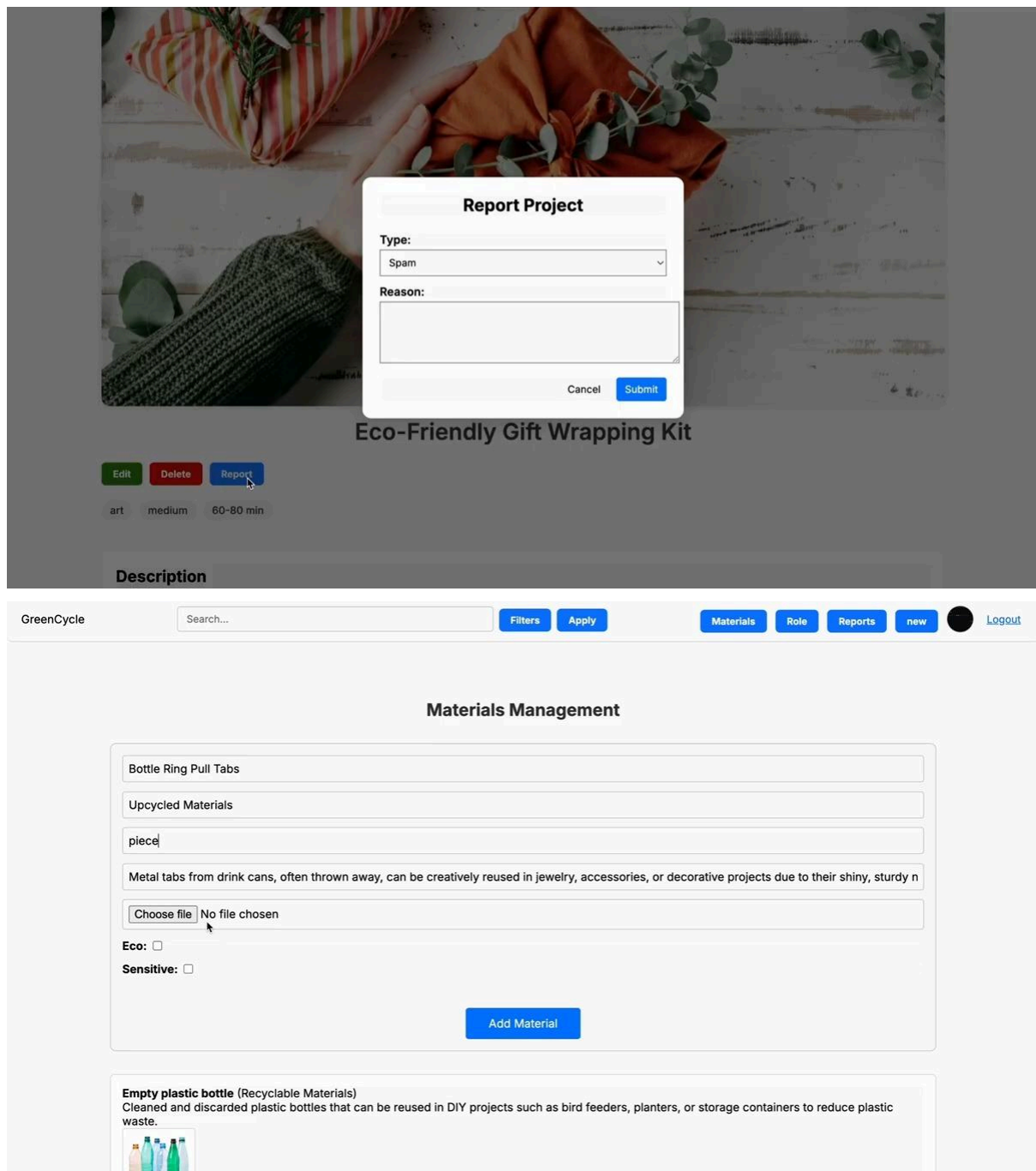
Category: art
Difficulty: easy
Time: 30-45 min

Tin Can Herb Garden



Newspaper Gift Bags





10. Team Contribution Assessment

Boris Marinković: Product Owner, documentation author, and frontend developer. Led requirements definition, wrote the project charter and other project documentation, contributed to frontend development, and assisted with backend implementation and database-related work.

Janko Kondić: Scrum Master and backend developer. Managed sprint coordination and backlog, set up the Node.js/Express backend and database connection,

implemented backend functionality including authentication and role-based access control, managed deployment, and resolved technical blockers during development.

Nikola Gostovikj: Full-stack support developer and tester. Assisted with frontend and backend development, contributed to project structure and implementation, connected frontend and backend components, and facilitated usability testing sessions.

Work was distributed according to roles and actual skills rather than task count, a decision established in Sprint 0 and maintained throughout the project.

11. Conclusions

Looking back at the project, the most valuable research outcomes were not the confirmed hypotheses, but the partial confirmation and the usability issues discovered during testing. The finding that small eco-business owners would benefit from the platform only if key features functioned effectively provided a more realistic conclusion, while the usability problems in Task 3 directly confirmed concerns identified earlier in the stakeholder analysis.

The engineering management decisions were also driven by practical needs rather than theory. Tasks were assigned according to skills instead of equal distribution to minimize project risk in a small four-month development environment.

Although Sprint 3 and Sprint 4 are still unfinished and some usability issues remain unresolved, the project now has a solid technical and methodological foundation with clearly defined next steps.